

```
# COMPLETE DATA ANALYSIS (AND ASSIGNMENT)
```

```
# ----- 0. SETUP -----
```

```
library(dplyr)
```

```
library(visdat)
```

```
library(naniar)
```

```
library(mice)
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
library(dbscan)
```

```
# File path (adjust as needed)
```

```
data_path <- "C:/Users/pc/Desktop/TANKOFOLDER/DataSet_No_Details.csv"
```

```
# ----- 1. READ & CLEAN DATA -----
```

```
df <- read.csv(data_path)
```

```
# Remove columns not needed
```

```
cols_to_remove <- c("h_index_34", "h_index_56", "hormone10_1", "hormone10_2",
```

```
  "an_index_23", "outcome", "factor_eth", "factor_h",
```

```
  "factor_pcos", "factor_prl")
```

```
MD_df <- df %>% select(-any_of(cols_to_remove))
```

```
# Keep factor variables separately (optional)
```

```
factor_df <- df %>% select(record_id, outcome, factor_eth, factor_h,
```

```
  factor_pcos, factor_prl)
```

```
# ----- 2. MISSING DATA ANALYSIS -----
```

```
total_na <- sum(is.na(MD_df))
```

```
na_per_col <- colSums(is.na(MD_df))
```

```
na_percent <- colMeans(is.na(MD_df)) * 100
```

```
cat("\n=== Missing Data Overview ===\n")
```

```
cat("Total NAs:", total_na, "\n")
```

```
print(na_per_col[na_per_col > 0])
```

```
# Columns with >35% missing – will be removed
```

```
high_miss <- names(na_percent[na_percent > 35])
```

```
cat("\nColumns with >35% missing:\n"); print(high_miss)
```

```
# Visualise missing patterns
```

```
vis_miss(MD_df)
```

```
gg_miss_var(MD_df)
```

```
# Remove high- missing columns
```

```
cols_to_remove1 <- c("hormone9", "hormone11", "hormone12", "hormone13", "hormone14")
```

```
handle_MD_df <- MD_df %>% select(-any_of(cols_to_remove1))
```

```
# ----- 3. LITTLE'S MCAR TEST (HOMEWORK) -----
```

```
numeric_data <- handle_MD_df %>% select(where(is.numeric))
```

```
mcár_result <- mcár_test(numeric_data)
```

```
p_value <- mcar_result$p.value
```

```
cat("\n=== Little's MCAR Test ===\n")
```

```
print(mcar_result)
```

```
cat("\nInterpretation:\n")
```

```
if (p_value > 0.05) {
```

```
  cat("p =", p_value, "> 0.05 → Data is MCAR. Multiple imputation is suitable.\n")
```

```
} else {
```

```
  cat("p =", p_value, "≤ 0.05 → Data is NOT MCAR. Use advanced methods.\n")
```

```
}
```

```
# ----- 4. MULTIPLE IMPUTATION (PMM) -----
```

```
imputed_object <- mice(handle_MD_df, m = 5, method = "pmm", print = FALSE)
```

```
imputed_full <- complete(imputed_object)
```

```
# Remove any rows with remaining NAs (required for LOF)
```

```
imputed_clean <- na.omit(imputed_full)
```

```
cat("\nRows removed due to NAs:", nrow(imputed_full) - nrow(imputed_clean), "\n")
```

```
# ----- 5. DENSITY PLOT (ORIGINAL vs IMPUTED) -----
```

```
if ("hormone10_generated" %in% names(handle_MD_df)) {
```

```
  ggplot() +
```

```
    geom_density(data = handle_MD_df, aes(x = hormone10_generated, fill = "Original"),
```

```
               alpha = 0.5, na.rm = TRUE) +
```

```
    geom_density(data = imputed_clean, aes(x = hormone10_generated, fill = "Imputed"),
```

```

      alpha = 0.5, na.rm = TRUE) +

labs(title = "Density: Original vs Imputed (hormone10_generated)") +

scale_x_continuous(limits = c(0,2)) +

theme_minimal()
} else {

first_num <- names(handle_MD_df)[sapply(handle_MD_df, is.numeric)][1]

rbind(data.frame(value = handle_MD_df[[first_num]], Method = "Original"),

      data.frame(value = imputed_clean[[first_num]], Method = "Imputed")) %>%

ggplot(aes(x = value, fill = Method)) +

geom_density(alpha = 0.5, na.rm = TRUE) +

labs(title = paste("Density:", first_num, "- Original vs Imputed")) +

theme_minimal()

}

```

```

# ----- 6. TRADITIONAL OUTLIER BOXPLOTS -----

```

```

# Lipid variables

```

```

lipid_cols <- c("lipids1", "lipids2", "lipids3", "lipids4", "lipids5")

lipid_present <- lipid_cols[lipid_cols %in% names(imputed_clean)]

if (length(lipid_present) > 0) {

  imputed_clean %>%

  select(all_of(lipid_present)) %>%

  pivot_longer(everything(), names_to = "var", values_to = "val") %>%

  ggplot(aes(x = var, y = val)) +

  geom_boxplot(fill = "lightblue", alpha = 0.7) +

  labs(title = "Outlier Detection - Lipid Variables") +

```

```

    theme_minimal()
  }

# All numeric variables

imputed_clean %>%

  select(where(is.numeric)) %>%

  pivot_longer(everything(), names_to = "var", values_to = "val") %>%

  ggplot(aes(y = val)) +

  geom_boxplot(fill = "steelblue", alpha = 0.6, outlier.color = "red") +

  facet_wrap(~var, scales = "free", ncol = 4) +

  labs(title = "Boxplots for All Numeric Variables") +

  theme_minimal() +

  theme(axis.text.x = element_blank())

# ----- 7. LOCAL OUTLIER FACTOR (LOF) -----

lof_dataset <- imputed_clean

lof_data <- lof_dataset %>% select(where(is.numeric))

# Remove constant columns

constant <- sapply(lof_data, function(x) sd(x, na.rm = TRUE) == 0)

if (any(constant)) lof_data <- lof_data %>% select(-which(constant))

# Scale

lof_scaled <- scale(lof_data)

```

```

# LOF parameters

minPts_val <- min(20, floor(nrow(lof_scaled) / 5))

cat("\nLOF using minPts =", minPts_val, "\n")


# Compute LOF

lof_scores <- lof(lof_scaled, minPts = minPts_val)


# Add scores and outlier flag

lof_dataset$LOF_Score <- lof_scores

threshold <- 1.5

lof_dataset$Is_Outlier <- lof_scores > threshold


# Summary

cat("\nLOF Score Statistics:\n")

print(summary(lof_scores))

cat("\nOutliers detected:", sum(lof_dataset$Is_Outlier),

    "(", round(mean(lof_dataset$Is_Outlier)*100, 2), "%)\n")


# Top 10 outliers

cat("\nTop 10 outliers by LOF score:\n")

lof_dataset %>%

  arrange(desc(LOF_Score)) %>%

  select(where(is.numeric), LOF_Score, Is_Outlier) %>%

  head(10) %>%

  print()

```

```
# ----- 8. LOF VISUALISATIONS -----
```

```
# Histogram
```

```
ggplot(lof_dataset, aes(x = LOF_Score)) +  
  geom_histogram(aes(fill = after_stat(count)), bins = 40, color = "black") +  
  geom_vline(xintercept = threshold, color = "red", linetype = "dashed", size = 1) +  
  geom_vline(xintercept = 1, color = "blue", linetype = "dotted") +  
  scale_fill_gradient(low = "lightblue", high = "darkblue") +  
  labs(title = "LOF Distribution",  
       subtitle = paste("minPts =", minPts_val, " | ", sum(lof_dataset$Is_Outlier), "outliers")) +  
  theme_minimal()
```

```
# Bivariate scatterplot (first two numeric variables)
```

```
if (ncol(lof_data) >= 2) {  
  var1 <- names(lof_data)[1]  
  var2 <- names(lof_data)[2]  
  ggplot(lof_dataset, aes(x = .data[[var1]], y = .data[[var2]])) +  
    geom_point(aes(color = Is_Outlier, size = LOF_Score), alpha = 0.6) +  
    scale_color_manual(values = c("FALSE" = "blue", "TRUE" = "red")) +  
    labs(title = "Bivariate Scatterplot with LOF Outliers") +  
    theme_minimal()  
}
```

```
# Boxplot of LOF scores by outlier status
```

```
ggplot(lof_dataset, aes(x = Is_Outlier, y = LOF_Score, fill = Is_Outlier)) +
```

```

geom_boxplot(alpha = 0.7) +
geom_hline(yintercept = threshold, linetype = "dashed", color = "darkred") +
scale_fill_manual(values = c("FALSE" = "lightblue", "TRUE" = "red")) +
labs(title = "LOF Score: Normal vs Outlier") +
theme_minimal()

```

```

# ----- 9. FINAL SUMMARY -----

```

```

cat("\n===== FINAL SUMMARY =====\n")

cat("1. Little's MCAR Test: p =", p_value,
    ifelse(p_value > 0.05, "→ Data is MCAR", "→ Data is NOT MCAR"), "\n")

cat("2. Imputation: PMM, original NAs =", sum(is.na(handle_MD_df)),
    ", after cleaning =", sum(is.na(imputed_clean)), "\n")

cat("3. LOF: minPts =", minPts_val, ", outliers =", sum(lof_dataset$Is_Outlier),
    "(", round(mean(lof_dataset$Is_Outlier)*100, 2), "%)\n")

cat("4. LOF score range:", round(min(lof_scores), 3), "–", round(max(lof_scores), 3), "\n")

cat("\nAnalysis completed successfully.\n")

```